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INTRODUCTION TO MACHINE LEARNING

At its core, machine learning (ML) is the study of algorithms that improve automatically through experience. Unlike traditional computer programs, which follow explicit instructions, ML models **learn patterns from data** and generalise to unseen situations.

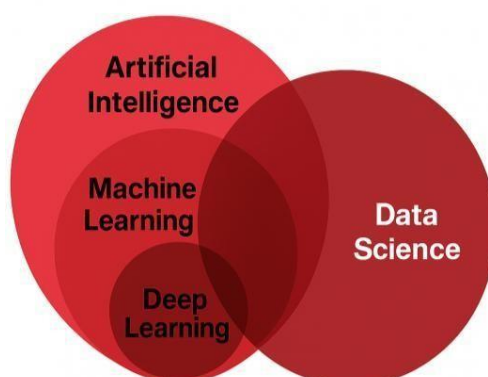
Arthur Samuel (1959), one of the pioneers of the field, defined machine learning as “*the field of study that gives computers the ability to learn without being explicitly programmed.*”

Machine Learning in the Landscape of AI and Data Science

ML sits within a broader technological ecosystem:

- **Artificial Intelligence (AI):** The broader discipline concerned with making machines behave intelligently. ML is a major approach within AI.
- **Data Science (DS):** A multidisciplinary field combining statistics, computer science, and domain knowledge. ML is one of the main tools of data science, especially for predictive analytics.
- **Deep Learning (DL):** A subfield of ML based on artificial neural networks with many layers. DL powers breakthroughs in speech recognition, image analysis, and large language models.

You may visualise the relations between these fields as follows:

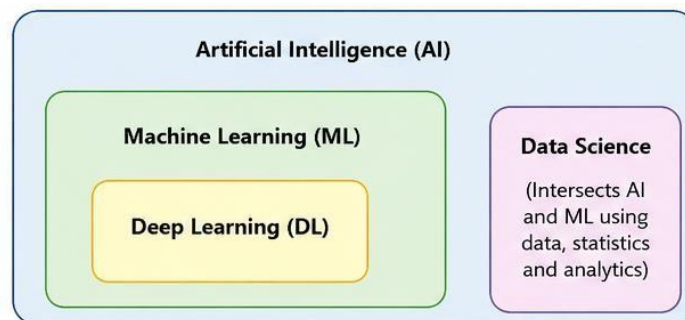


Although Artificial Intelligence, Machine Learning, Deep Learning, and Data Science are closely related, they are not interchangeable. Each serves a different purpose while often working together in real-world applications.

Technology	Primary Goal	Learns from Data?	Example
Artificial Intelligence	Mimic intelligent behaviour	Sometimes	Chess-playing system
Machine Learning	Learn patterns from data	✓	House price prediction
Deep Learning	Learn complex representations	✓	Face recognition
Data Science	Extract insights from data	Optional	Sales trend analysis

Remember: Every Deep Learning model is a Machine Learning model, but not every Machine Learning model is a Deep Learning model.

Relationship between Artificial Intelligence, Machine Learning, Deep Learning, and Data Science can be visualised below.



Solved Example:

Identifying the Appropriate Technology

A retail company wants to build different intelligent systems.

Requirement	Suitable Technology
Predict next month's sales	Machine Learning
Detect objects from CCTV footage	Deep Learning

Requirement	Suitable Technology
Create an interactive chatbot	Artificial Intelligence
Analyse yearly sales trends	Data Science

Explanation:

- Predicting future sales involves learning from historical data, making Machine Learning the appropriate choice.
- Object detection requires extracting complex visual features from images. Hence, Deep Learning is preferred.
- Analysing sales reports focuses on understanding historical data rather than building predictive models, making it a Data Science task.

Evolution of Machine Learning

Understanding the history of ML helps place today's trends in perspective.

- **1G50s-1G60s (Foundations):** Early ideas of perceptrons, symbolic AI, and Samuel's checkers-playing program
- **1G70s-1G80s (Knowledge-based systems):** Expert systems dominated AI, but ML research continued with decision trees and Bayesian methods
- **1G90s-2000s (Statistical ML):** Support Vector Machines (SVM), ensemble methods (bagging, boosting), and probabilistic graphical models rose to prominence
- **2010s (Deep Learning era):** Neural networks scaled with massive data and GPU computing, leading to breakthroughs in vision (ImageNet, 2012) and language (Word2Vec, 2013)
- **2020s (Foundation Models):** Large-scale pretraining produced LLMs like GPT, Gemini, and Claude. Yet, classical ML still plays a vital role in industry and research

Timeline Summary

Time Period	Major Development	Real-World Impact
1950–1960	Perceptrons and Symbolic AI	First intelligent programs
1970–1980	Expert Systems	Rule-based decision making
1990–2000	Statistical Machine Learning	Better prediction models
2010–2020	Deep Learning	Image recognition, NLP
2020–Present	Foundation Models	ChatGPT, Gemini, Claude

Why Should We Learn Classical Machine Learning?

Although Large Language Models have become extremely popular, classical machine learning remains the preferred choice for many business problems involving structured datasets because it is easier to interpret, requires less computational power, and often provides competitive predictive performance.

Key Paradigms of Machine Learning

Machine Learning (ML) is not a single unified approach. Instead, it comprises **different paradigms**, each defined by how data is presented to the learner and what kind of feedback is available. These paradigms represent **philosophies of learning**, shaping the choice of algorithms and applications.

The five principal paradigms are:

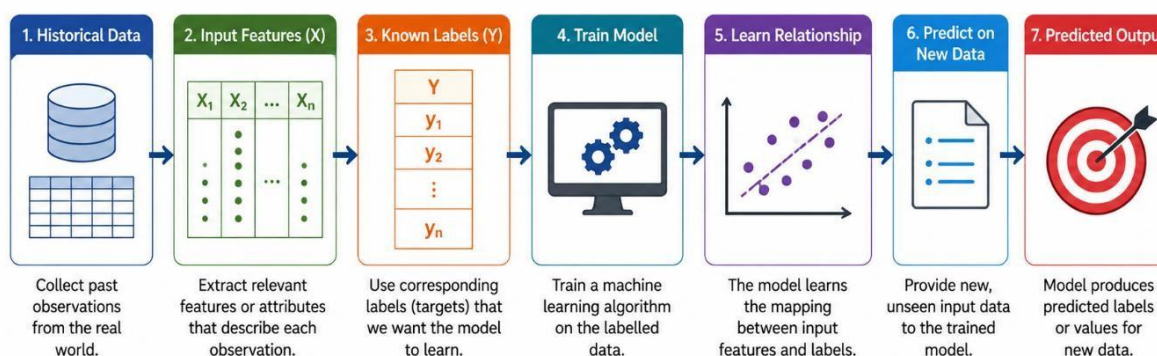
1. **Supervised Learning**
2. **Unsupervised Learning**

3. Semi-Supervised Learning
4. Reinforcement Learning
5. Self-Supervised Learning

Supervised Learning

- **Definition:** Supervised learning refers to training a model on a **labelled dataset**, where each input x has a corresponding output y . The task is to learn a function $f(x)$ that approximates the mapping between inputs and outputs.
- **Mathematical Formulation:** Learn a function $f : x \rightarrow y$ that minimises prediction error.
- **Intuition:** Imagine a student learning arithmetic from a teacher. The teacher provides many worked examples (input $\rightarrow$ correct answer). Over time, the student learns to generalise to new problems.
- Similarly, supervised learning models learn rules from examples to predict unseen cases.

Supervised Learning Workflow



Types of Supervised Learning Problems

1. **Regression** – Predict continuous outputs
 - Algorithms: Linear Regression, Ridge/Lasso Regression, Neural Networks
 - Loss Function: Mean Squared Error (MSE)

- Example: Predicting house prices from features such as size, location, and number of bedrooms

2. **Classification** – Predict discrete class labels

- Algorithms: Logistic Regression, Decision Trees, Support Vector Machines (SVM), k-NN
- Loss Function: Cross-Entropy Loss
- Example: Email classification (spam / not spam)

Note: Despite its name, logistic regression is a classification algorithm: it uses a regression-like equation to model the log-odds, then applies the sigmoid function to produce probabilities that are thresholded into class labels.

Applications of Supervised Learning

- Predictive maintenance in manufacturing
- Disease diagnosis from medical scans
- Credit scoring in banking
- Image recognition, speech recognition

Advantages of Supervised Learning

- Provides high accuracy when labelled data is abundant
- Well-studied theory, with many established algorithms

Limitations of Supervised Learning

- Requires large amounts of **labelled data** (expensive)
- Can overfit if not regularised properly

Unsupervised Learning

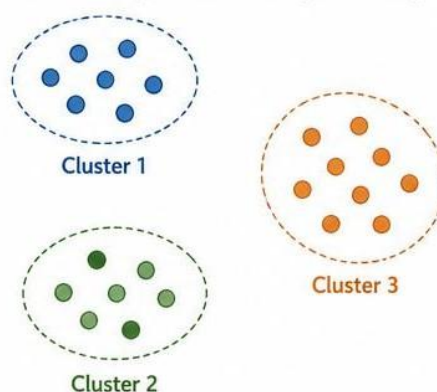
- **Definition:** Unsupervised learning deals with **unlabelled data**, aiming to uncover **hidden patterns or structures** without explicit guidance
- **Intuition:** Think of an archaeologist sorting ancient pottery with no knowledge of their origin. By comparing shapes, textures, and decorations, the archaeologist clusters similar items together. This is analogous to clustering in ML

Types of Unsupervised Learning Problems

1. **Clustering:** Group data into clusters such that items within a cluster are more like each other than to those in other clusters
 - Algorithms: K-Means, Hierarchical Clustering, DBSCAN (Density-Based Spatial Clustering of Applications with Noise)
2. **Dimensionality Reduction:** Represent data in fewer dimensions while preserving essential structure
 - Algorithms: Principal Component Analysis (PCA), t-SNE (t-distributed Stochastic Neighbour Embedding), Autoencoders

Clustering groups similar observations without requiring labelled data can be visualised below.

Customer Segmentation using Clustering



Applications of Unsupervised Learning

- Customer segmentation in marketing
- Anomaly detection (e.g., fraud detection)
- Visualisation of high-dimensional datasets
- Topic modelling in text corpora

Advantages of Unsupervised Learning

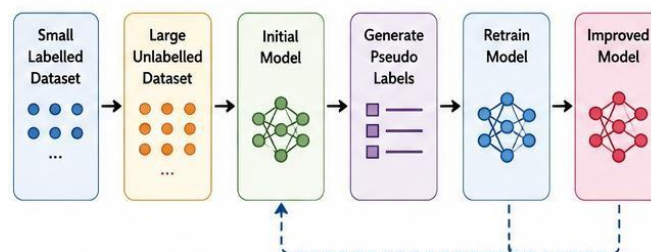
- Does not require labelled data
- Can reveal hidden structures humans may not notice

Limitations of Unsupervised Learning

- Interpretability of clusters can be subjective
- No standard measure of success as the “ground truth” often missing

Semi-Supervised Learning

- **Definition:** Semi-supervised learning lies between supervised and unsupervised paradigms. It uses a **small amount of labelled data** and a **large pool of unlabelled data** to improve learning.



- **Intuition:** Imagine trying to learn a new language with only a handful of translated sentences (labelled) and a massive book in that language without translations (unlabelled). You leverage the few examples to make sense of the rest.

Approaches for Semi-Supervised Learning

1. **Self-Training:** Train a model on labelled data, predict labels for unlabelled data, and iteratively.
2. **Graph-Based Methods:** Represent data points as nodes in a graph, propagating labels through edges.
3. **Consistency Regularisation:** Encourage consistent predictions under perturbations (e.g., noisy inputs).

Applications of Semi-Supervised Learning

- Medical imaging (where labelling requires experts)
- Web content classification
- Fraud detection in financial datasets

Advantages of Semi-Supervised Learning

- Reduces the need for expensive labelling
- Often achieves accuracy close to supervised learning with less data

Limitations of Semi-Supervised Learning

- Risk of reinforcing errors if pseudo-labels are incorrect
- Requires careful design of assumptions about the data

Reinforcement Learning

- **Definition:** Reinforcement Learning (RL) is a paradigm where an **agent learns to act** in an environment by **trial and error**, receiving **rewards or penalties** as feedback

Interaction cycle in reinforcement learning can be visualised by the figure below.



Applications of Reinforcement Learning

- Robotics (autonomous navigation, manipulation)
- Games (Atari, Chess, Go, StarCraft)
- Recommendation systems (personalised policies)
- Finance (portfolio management)

Advantages of Reinforcement Learning

- Can handle sequential decision-making problems
- Learns optimal strategies without explicit supervision

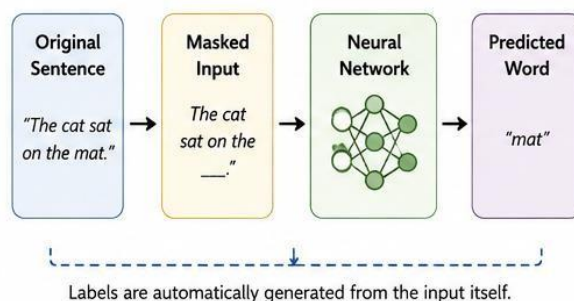
Limitations of Reinforcement Learning

- Sample inefficiency (requires many trials)
- Stability and convergence are challenging

Self-Supervised Learning

- **Definition:** Self-supervised learning is a paradigm where a model teaches itself using the raw data. Instead of relying on external labels, it creates its own training signals by hiding part of the data and trying to predict it.

Self-supervised learning creates labels directly from the input data.



- **Intuition:** Imagine solving a crossword puzzle: you don't have a teacher telling you the answers, but you use the surrounding letters (context) to guess the missing word

Applications of Self-Supervised Learning

- **Natural Language Processing (NLP):** Foundation of LLMs (ChatGPT, Gemini, etc.)
- **Computer Vision (CV):** Self-supervised pretraining on large imagedatasets
- **Speech Recognition:** Predicting masked audio segments

Advantages of Semi-Supervised Learning

- Harnesses vast unlabelled data
- Produces general-purpose representations that transfer across tasks

Limitations of Semi-Supervised Learning

- Computationally expensive
- Risk of capturing spurious correlations without careful design

Paradigm	Input Data	Goal	Example Use Case	Key Limitation
Supervised Learning	Labelled	Predict outputs	Spam detection	Needs labelled data

Unsupervised Learning	Unlabelled	Discover patterns	Customer segmentation	Hard to evaluate results
Semi-Supervised	Few labels + unlabelled	Improve learning with limited labels	Medical imaging	Error propagation
Reinforcement Learning	Interaction data	Maximise cumulative reward	AlphaStar, Robotics	Sample inefficient
Self-Supervised	Raw data (self-labelled)	Learn representations	LLMs, image pretraining	Requires massive compute

Paradigm	Supervised Learning	Unsupervised Learning	Semi-Supervised Learning	Reinforcement Learning	Self-Supervised Learning
Data	Labelled data (X, Y)	Unlabelled data (X)	Few labelled (Y) + Many unlabelled (X)	Interaction data (state, action, reward)	Unlabelled data (X) with self-generated labels
Goal	Learn mapping from X to Y	Discover hidden structure or patterns	Improve learning using additional unlabelled data	Learn policy to maximize cumulative reward	Learn meaningful representations from data
Feedback	Explicit labels provided	No labels	Partial labels available	Reward signal	Automatically generated labels
Typical Tasks	Classification, Regression	Clustering, Association, Dimensionality Reduction	Text classification, Image classification with limited labels	Game playing, Robotics, Resource allocation	Pretraining models, Representation learning
Examples of Algorithms	Linear Regression, SVM, Decision Trees	K-Means, PCA, Apriori	Label Propagation, Pseudo-Labeling	Q-Learning, Deep Q-Networks, Policy Gradient	Masked Language Models, Contrastive Learning

Why Classical ML Still Matters

The emergence of deep learning and large-scale models has revolutionised artificial intelligence. Nevertheless, classical machine learning techniques continue to play an indispensable role in both academic study and industrial practice. Their relevance is sustained by the following considerations:

1. **Interpretability:** Techniques like logistic regression and decision trees are

explainable, i.e., you can tell how they make decisions and why a prediction was made

2. **Efficiency:** Require fewer computational resources
3. **Structured Data:** Still outperform deep models in many tabular datasets
4. **Educational Value:** Concepts like generalisation, overfitting, and optimisation originate here

Challenges in Machine Learning

While machine learning has enabled transformative advances, its use is accompanied by important challenges. These challenges are both technical and ethical in nature.

1. Data Quality

Machine learning models are highly dependent on the quality of their training data. Datasets may contain missing values, noise, or biases, which degrade the reliability of predictions.

- Missing, noisy, or biased data → unreliable models
- Example: Biased facial recognition datasets leading to unfair outcomes

2. Generalisation and Overfitting

A central goal in machine learning is **generalisation** — the ability to perform well on unseen data. A common challenge is **overfitting**, where the model performs exceptionally well on training data but fails to generalise.

- **Overfitting:** Low training error, high test error
- Solutions: Regularisation, cross-validation, dropout, early stopping

3. Bias, Fairness, and Ethics

ML models inevitably reflect the data they are trained on. If the data contains biases, the model may amplify them. This raises serious issues of fairness, accountability, and ethical deployment.

- Models amplify biases in data
- Ethical frameworks (e.g., EU AI Act) stress fairness and accountability

4. Interpretability

ML models inevitably reflect the data they are trained on. If the data contains biases, the model may amplify them. This raises serious issues of fairness, accountability, and ethical deployment.

- Black-box models raise trust issues
- Explainable AI (XAI): SHAP, LIME, attention maps

Applications of ML

Machine learning has become ubiquitous across domains, driving innovation in multiple sectors:

- **Healthcare:** Cancer detection from scans, personalised medicine
- **Finance:** Fraud detection, algorithmic trading, credit scoring
- **Retail:** Recommendation engines (Amazon, Netflix, etc.)
- **Transportation:** Autonomous vehicles, traffic optimisation
- **Language and Vision:** Speech assistants, translation, object recognition

The Future of Machine Learning

The field of machine learning continues to evolve rapidly, with several emerging directions shaping its trajectory:

- **Responsible AI:** Fairness, transparency, and ethics
- **Green AI:** Energy-efficient models
- **Neuro-symbolic AI:** Combining ML with symbolic reasoning
- **AutoML:** Democratisation of ML tools

- Human-AI Collaboration: AI as decision support, not replacement

Date	Stock Volume	Value (\$)
20250820	209,456	14.56
20250825	223,198	15.46
20250914	199,097	13.89
20251004	234,435	14.67
20251113	222,987	18.97

Linear Regression

Introduction to Linear Regression

Linear regression is one of the most fundamental and widely used techniques in data analysis and machine learning. It is a supervised algorithm that predicts a numerical output by modelling its relationship with input variables as a linear function. This makes it a natural starting point for understanding more advanced algorithms.

The central idea of regression is to define how one or more input variables (called predictors or independent variables) relate to an output variable (the dependent variable). By studying these relationships, analysts can identify meaningful patterns in data and use them to make predictions about future outcomes.

At its core, linear regression is essentially about **fitting the best possible straight line** through the data points. This line summarises the trend in the data, showing how changes in the input variables are associated with changes in the output variable, and provides a simple yet powerful way to model relationships.

A Simple Example

Suppose you are placing an order at your favourite bakery. You have often visited this bakery before and know how long it usually takes to prepare different numbers of muffins.

Can you predict how much time it will take to prepare a certain number of muffins today?

Here:

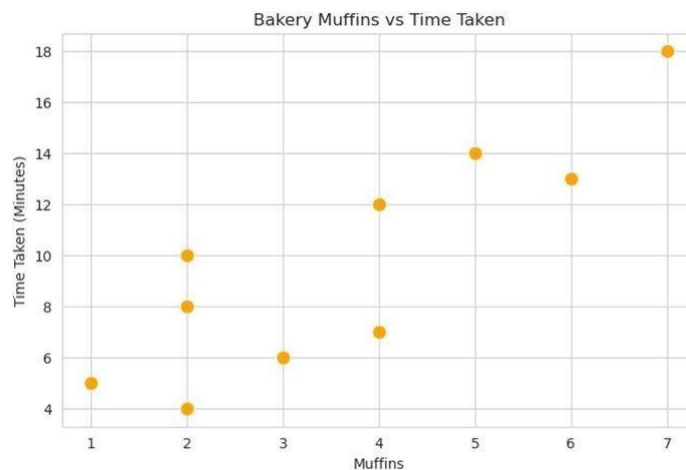
- The **input variable** (predictor) = number of muffins
- The **output variable** (target) = preparation time

Visualise the Data

From your past visits, you have gathered the following data for the time taken to deliver a certain number of muffins:

Number of Muffins (x)	Time Taken (minutes, y)
1	5
2	8
4	12
3	6
2	4
5	14
7	18
4	7
2	10
6	13

We can plot past data as points on a graph, with muffins on the X-axis and preparation time on the Y-axis. When we plot the above data, we notice a roughly straight-line pattern.



Each dot represents one observation. The upward trend suggests that time increases linearly with the number of muffins.

The Best Fit Line

Linear regression aims to draw a straight line that **best fits** the data. In other words, we want a regression line that makes our prediction of the output as close as possible to the actual value of the output for a given input.

The general equation of a line is

$$\hat{y} = \beta_0 + \beta_1 x_1$$

Here,

- $\hat{y}$ = predicted value(e.g., preparation time)
- x = input variable(e.g., number of muffins)
- β_0 = intercept (base preparation time when muffins = 0)
- β_1 = slope of the line (extra time required per muffin)

There are many ways in which we could find our best fit line; the line will be considered the “best fit” if the predictions are as close as possible to the actual values.

Solved Example 1

Predicting Preparation Time Using a Linear Regression Model

Suppose the bakery has already trained a linear regression model using historical order data. The resulting regression equation is

$$\hat{y} = 2.60 + 1.97x$$

where

- x = Number of muffins ordered
- $\hat{y}$ = Estimated preparation time (minutes)

A customer places an order for **8 muffins**.

Estimate the preparation time using the regression model.

Solution

Substituting $x = 8$,

$$\hat{y} = 2.60 + 1.97(8)$$

$$\hat{y} = 2.60 + 15.76$$

$$\hat{y} = 18.36 \text{ minutes}$$

Therefore, the bakery should expect the preparation time to be approximately **18.4 minutes**.

Interpretation

This value represents the model's prediction based on historical observations. The actual preparation time may differ slightly because of operational factors such as staff availability, oven occupancy, or simultaneous customer orders.

Errors and the Loss Function

In machine learning, an **error** is the difference between a model's prediction and the actual, or "ground truth," value, while a **loss function** (or cost/error function) is a mathematical tool that quantifies this error by assigning a numerical "cost" to it.

The loss function's primary role is to provide feedback during a model's training process, guiding it to minimise the quantified error and thereby improve its predictions. A lower loss value signifies a more accurate model, while a higher loss indicates greater inaccuracy and a need for adjustment.

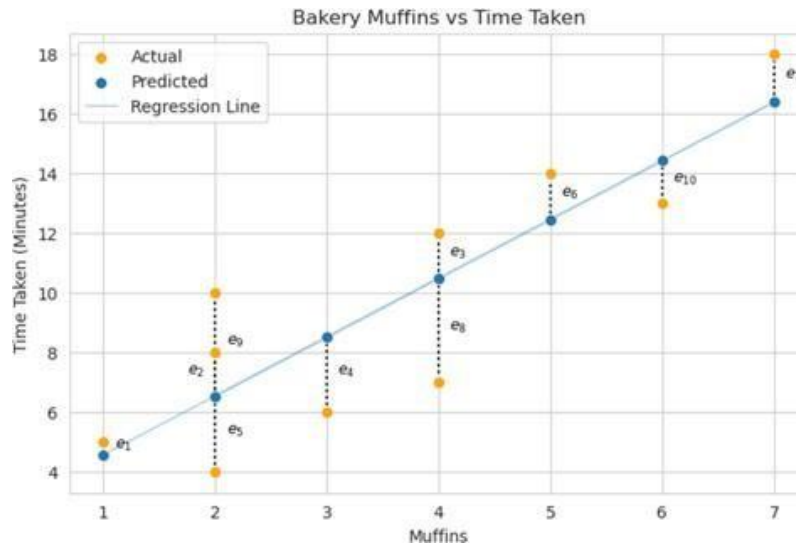
When we fit a line through the muffin data, not all points lie exactly on the line. The difference between the observed value and the predicted value is called the **error** or **residual**.

We can denote the error in prediction for a point as:

$$e_i = y_i - \hat{y}_i$$

Here,

- e_i = error or residual
- y_i = observed value
- $\hat{y}_i$ = predicted value



Now, as we will have different sets of lines contesting to be the most optimal one, we will have different errors for each point. For one line, the error for point A can be lower whereas for point B it can be much higher. How do we get to a single measure of error loss for each line?

One basic way is to simply add the errors. But this will lead to the positive and negative errors cancelling out each other. The error values can be positive or negative depending on where the true value falls with respect to the regression line.

Analytical Example: Why Residuals Cannot Be Added Directly

Consider two candidate regression models developed for the bakery's order preparation system.

Model A

Order	Residual (minutes)
1	+4
2	-4
3	+3
4	-3

Total Error

$$(+4) + (-4) + (+3) + (-3) = 0$$

Model B

Order	Residual (minutes)
1	+1
2	-1
3	+1
4	-1

Total Error

$$(+1) + (-1) + (+1) + (-1) = 0$$

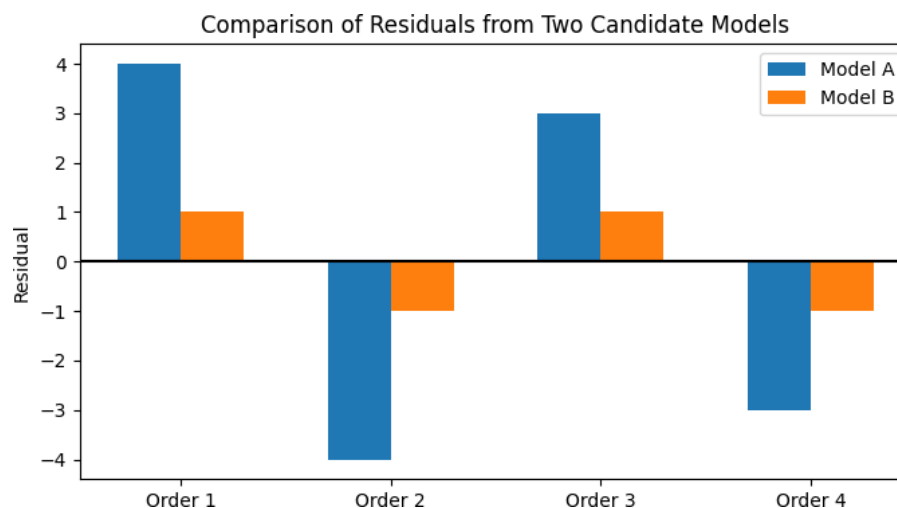
If residuals were simply summed, **both models would appear equally accurate**, because their total error is zero.

However, Model A makes substantially larger prediction errors than Model B. Therefore, directly adding residuals provides a misleading measure of model performance.

To overcome this limitation, regression models square each residual before aggregation, ensuring that:

- Positive and negative errors do not cancel each other.
- Larger prediction errors receive proportionally greater penalty.
- A single objective function can be minimised during model training.

This motivates the use of the **Residual Sum of Squares (RSS)** and the **Mean Squared Error (MSE)** introduced next.



Although the residuals of both models sum to zero, Model A produces considerably larger

prediction errors than Model B, motivating the use of squared-error metrics.

To avoid this, we can **square** the errors

- Squaring makes all errors positive
- Larger errors are penalised more than smaller ones

A commonly used squared error metric is the **mean squared error (MSE)**, obtained by taking the mean of these squared errors:

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n e_i^2 = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

Here,

- y_i = observed value
- $\hat{y}_i = \beta_0 + \beta_1 x_1$ = predicted value
- n = number of observations

Why use MSE?

- It gives a **single number** that tells us how well the regression line fits the data
- A smaller MSE means the predictions are closer to the actual values which means better fit
- It is smooth and differentiable, which makes it easier for mathematical optimisation

The regression line is the one that **minimises the MSE**, making it the best possible fit for the data.

Solved Example: Evaluating Model Performance for Bakery Operations

Scenario:

The bakery has deployed its linear regression model to estimate preparation time during peak business hours. After one week, the operations manager compares the model's predictions with the actual preparation times to evaluate whether the model is sufficiently accurate for production use.

Order	Muffins	Actual Time (min)	Predicted Time (min)
1	6	15.0	14.6
2	9	20.5	19.8
3	12	25.2	24.7
4	15	31.8	30.9
5	18	36.4	37.1

Analysis:

Rather than evaluating individual predictions in isolation, the bakery wants to understand whether the model consistently produces acceptable estimates for operational planning.

The prediction errors are summarised below.

Order	Residual (Actual – Predicted)
1	0.4
2	0.7
3	0.5
4	0.9
5	-0.7

The residuals are relatively small and are distributed on both sides of zero, indicating that the model does not consistently overestimate or underestimate preparation time. This behaviour suggests that the fitted regression model is suitable for forecasting bakery preparation time under normal operating conditions.

Business Insight:

In practice, data scientists rarely judge a regression model using a single prediction. They evaluate the distribution of residuals across many observations to determine whether the model is reliable enough for deployment.

Practice Exercise

The bakery data science team develops two competing regression models.

Metric	Model A	Model B
MSE	4.12	6.85
R ²	0.93	0.88

The operations manager asks which model should be deployed.

1. Compare the predictive performance of the two models.
2. Explain why MSE and R² should be interpreted together rather than independently.
3. Besides predictive accuracy, identify two operational considerations before deploying the model.

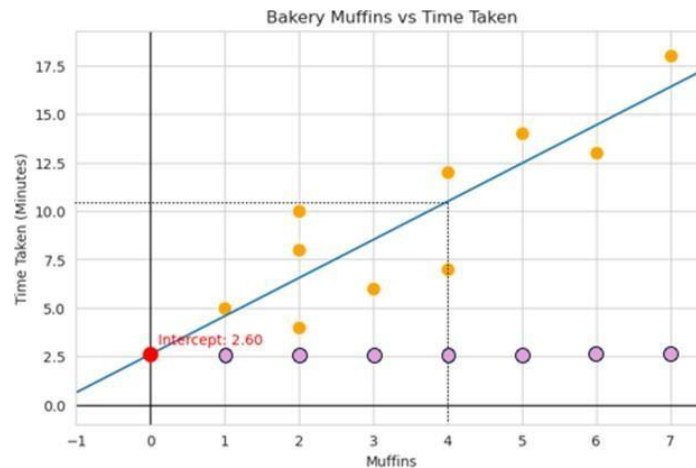
Interpreting the Equation

Suppose our bakery data gives the regression equation:

$$\hat{y} = 2.60 + 1.97x_1$$

This equation has two important parts:

- **Intercept** ($\beta_0 = 2.60$): The base preparation time, even with no muffins, is about 2.6 minutes
 - Always check the sensibility of your results depending on the business context. Does it really make sense to wait for no muffins?
- **Slope** ($\beta_1 = 1.97$): Each muffin adds about 1.97 minutes to the preparation time. For example, for 4 muffins, one needs to wait $2.60 + 1.97 \times 4 = 10.48$ minutes.



Worked Example: From Historical Data to a Prediction

The bakery manager wants to estimate preparation time for future orders using the historical observations collected earlier. Instead of manually drawing the best-fit line, the bakery decides to estimate the regression model using Python.

The objective is to learn the relationship between the **number of muffins ordered** and the **preparation time**, and then use the fitted model to predict preparation time for new customer orders.

Python Implementation

```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression

# Historical bakery data
bakery = pd.DataFrame({
    "Muffins": [1,2,4,3,2,5,7,4,2,6],
    "Preparation_Time": [5,8,12,6,4,14,18,7,10,13]
})

X = bakery[["Muffins"]]
y = bakery["Preparation_Time"]

model = LinearRegression()

model.fit(X, y)

print("Intercept:", round(model.intercept_,2))
```

```
print("Slope:", round(model.coef_[0],2))
```

Expected Output

Intercept: 2.60

Slope: 1.97

Interpretation

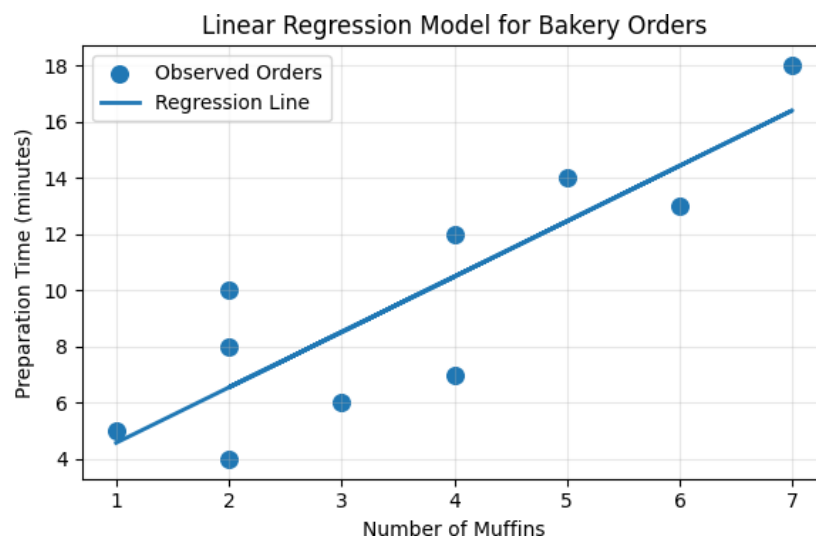
The fitted regression model is

$$\hat{y} = 2.60 + 1.97x$$

where

- **2.60** represents the baseline preparation time.
- **1.97** indicates that, on average, every additional muffin increases the preparation time by approximately **1.97 minutes**.

The values closely match the regression equation introduced earlier, illustrating how the coefficients are obtained directly from data rather than selected manually.



Although the regression line captures the overall trend, not every customer order falls exactly on the fitted line. Variations in staff availability, oven utilisation, order complexity, and concurrent customer requests introduce natural variability into preparation time. Consequently, the objective of linear regression is not to pass through every observation, but to estimate the relationship that minimises the overall prediction error across all observations.

Practice Exercise

The bakery has opened a second outlet and collected the following historical observations.

Muffins	Preparation Time (minutes)
3	8
5	13
7	16
9	21
11	24

Using the same workflow demonstrated above,

1. Fit a linear regression model using Python.
2. Report the intercept and slope.
3. Plot the fitted regression line.
4. Predict the preparation time for an order of **10 muffins**.
5. Compare the fitted model with the original bakery model and comment on possible operational differences between the two outlets.

Extending to Higher Dimensions

So far, we have only considered one predictor (number of muffins). But time taken may also depend on other factors — for example, the number of staff available in the bakery.

In that case, the regression model becomes:

$$\hat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_p x_p$$

Where:

- p = number of predictors
- β_i from 1 to p = coefficients of those respective predictors

In 3D, you fit a plane instead of a line, and in higher dimensions, the model extends this idea further. However, visualising beyond three dimensions becomes nearly impossible since human intuition is limited to 3D space.

Moreover, working with higher dimensions often brings extra challenges, such as **multicollinearity** and **heteroscedasticity**, which indicate that some of the core assumptions of linear regression are being violated.

Assumptions of Linear Regression

Linear regression, while simple and powerful, does not work well under all conditions. It has a set of **assumptions** that must hold true for the model to produce reliable results. These are:

- Linearity
- Independence of Data Points
- Homoscedasticity
- Normality of errors
- No multicollinearity

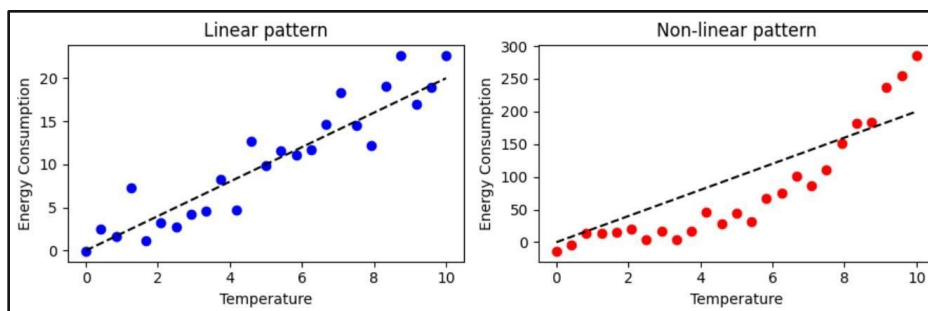
If these assumptions are ignored, predictions can become biased and/or misleading.

Variable relationships are rarely perfect, but understanding these assumptions allows us to judge when linear regression is appropriate.

Assumption: Linearity

The first assumption is that there is a **linear relationship** between the predictors and the target variable. This means that as the predictor changes, the target changes in a way that can be represented by a straight line.

If the relationship is **non-linear**, a straight line will fail to capture the true trend in the data.



Examples of Non-Linear Relationships

- **Population growth over time** → exponential curve (slow at first, then rapid)
- **Ocean tides** → periodic rise and fall due to gravitational pull of the moon and sun

Assumption: Independence of Data Points

The second assumption is that the data points (i.e., rows in the data) are **independent of each other**. This means that the value of one data point does not influence or depend on another.

If data points are interdependent, the model may pick up misleading patterns.

House ID	Square Footage	# Bedrooms	Price (\$)
115467	1,200	2	180,000
134245	1,500	3	220,000
214454	1,000	2	150,000
456565	1,800	4	270,000
342555	1,400	3	240,000

Here, data points (rows) do not influence each other (but the columns probably do; refer to the assumption of no multicollinearity to see how this might be a problem).

Date	Stock Volume	Value (\$)
20250820	209,456	14.56
20250825	223,198	15.46
20250914	199,097	13.89
20251004	234,435	14.67
20251113	222,987	18.97

Here, data points (rows) influence each other - today's stock volume and price depend on yesterday's.

Examples of Data with Dependence Between Observations

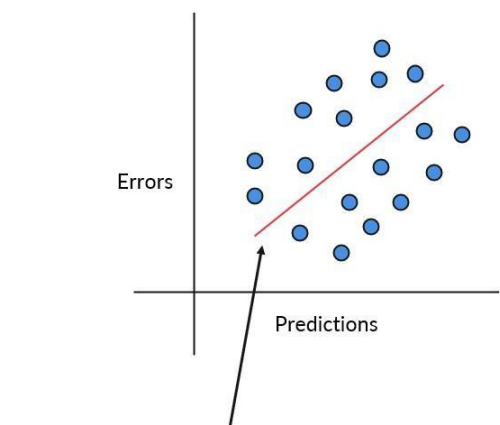
- In a network of sensors measuring pollution, readings from sensors placed close together often influence one another, as the pollution level in one area is similar to that in a neighbouring area
- In relative grading, your score may depend not only on your answers but also on how your peers perform

When data points are not independent, the conclusions drawn from linear regression become unreliable.

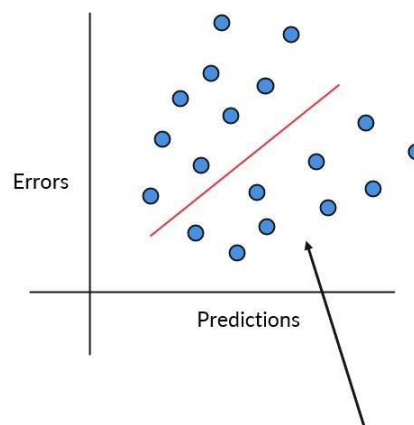
Assumption: Homoscedasticity of Errors

Another key assumption is **homoscedasticity**. This means that the variance of the errors (residuals) should remain constant across all levels of the predictor.

When this assumption is violated, we face **heteroscedasticity**, where the errors grow larger or smaller as the value of the predictor increases.



Homoscedastic (errors have constant variance)



Heteroscedastic (errors have varying variance)

In the left-hand plot, the data is homoscedastic, meaning the variance of the points stays roughly constant across the range. In contrast, the right-hand plot shows heteroscedasticity, where the spread of errors increases as the predictor values grow.

Examples of Homoscedastic Data

1. In household income and expenditure data, low-income families tend to have predictable and stable spending patterns. High-income households, however, display widely varying spending behaviours, creating heteroscedastic errors.
2. In delivery time data, local deliveries may always take roughly the same amount of time, while national or international deliveries vary greatly.

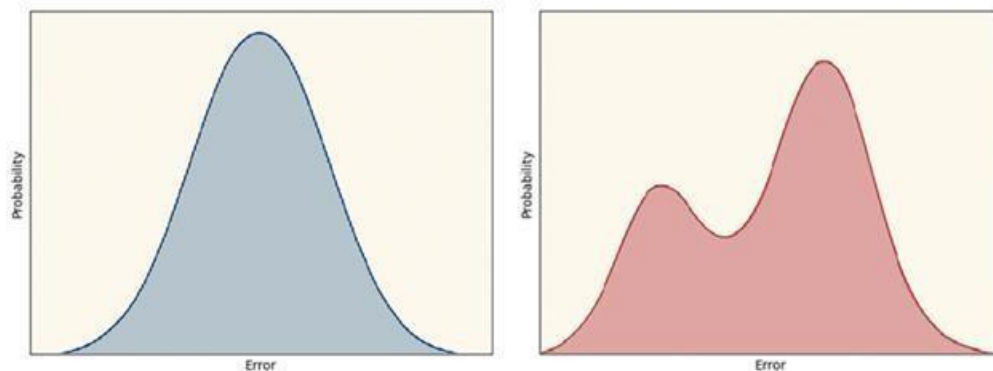
If errors are heteroscedastic, the reliability of predictions decreases because the model systematically underestimates or overestimates errors at certain ranges of the predictor.

Assumption: Normality of Errors

Linear regression also assumes that errors follow a **normal distribution** (the familiar bell-shaped curve).

If errors are normally distributed, most errors are small, with a few larger ones in either direction. This ensures that predictions are balanced and unbiased.

When errors deviate from normality, the regression results may not be suitable.



On the left, we have a normal distribution of errors. On the right, we can see two peaks - this is a multimodal distribution of errors where each peak seems normal.

Examples of Data with Non-Normally Distributed Errors

3. In medical insurance, smokers often pay much higher premiums than non-smokers, creating two distinct groups in the data. This clustering leads to non-normal errors.
4. In satellite image classification, predicting whether a region is water or land without distinguishing between forest, farmland, and cities can create sharp peaks in the error distribution

These cases show how non-normal errors signal that the linear regression model may not be appropriate.

Distinguishing Heteroscedasticity and Non-normal Errors

At first glance, heteroscedastic errors and non-normal errors may seem similar. However, they are different phenomena.

5. **Non-normal errors** occur when the distribution of errors is irregular, often with multiple peaks. For example, a pub may receive a very high number of orders on weekends compared to weekdays, creating two peaks in the error distribution.
6. **Heteroscedastic errors** occur when the variance of errors changes with the predictor. For example, folding a few clothes usually takes roughly the same time for everyone, but folding a large pile shows much more variability, creating heteroscedasticity.

Understanding the distinction helps in correctly diagnosing problems in data.

Assumption: No Multicollinearity

Finally, linear regression assumes that predictors (columns) are not highly correlated with each other. This is called the assumption of **no multicollinearity**.

When two predictors are strongly correlated, they carry nearly the same information. Keeping both adds little new predictive power but increases computational cost and may confuse the model.

	Cost of Item	Delivery Time	Volume of Item	Price
Cost of Item	1.000000	-0.134197	0.979261	0.836926
Delivery Time	-0.134197	1.000000	-0.135455	-0.019367
Volume of Item	0.979261	-0.135455	1.000000	0.820274
Price	0.836926	-0.019367	0.820274	1.000000

Here, we can see that the volume of the item and the cost of the item are highly correlated.

Examples of Data With Multicollinearity

7. In real estate data, house size (square footage) and number of bedrooms are highly correlated. Larger houses usually have more bedrooms, so including both variables introduces redundancy.
8. In air quality analysis, vehicle density and factory output often rise together in industrial cities, making them strongly correlated.

Multicollinearity reduces the interpretability of the regression model and weakens its ability to separate the true effect of each predictor.

Interpreting Regression Coefficients

Many models are more powerful than linear regression, but its true strength lies not just in making predictions, but in offering clear and meaningful interpretation of relationships between variables.

The intercept and coefficients in a regression model are not just numbers. They tell us about the baseline outcome and how each predictor influences the response. Understanding these values is crucial to connect our statistical model back to the real-world problem we are studying.

Interpreting the Intercept

The intercept in a regression equation is the predicted value of the response variable when all predictors are equal to zero. For example, if our bakery regression model is:

$$\text{Time} = 2.60 + 1.97x$$

The intercept (2.60) tells us that even if no muffins are ordered ($x = 0$), the bakery still takes about 2.6 minutes for preparation

- Note how, in this situation, this is not a meaningful result as one would not have to wait at all if no muffins are ordered.
- Always be aware of the business domain and problem at hand while interpreting coefficients or the intercept! Yet, when $x > 0$, this intercept term is meaningful. It represents the “base” order preparation time for your muffins.

In general, the intercept represents the **baseline** level of the response variable, before any predictors are considered.

Interpreting the Coefficients

Each coefficient tells us how much the response variable changes when the corresponding predictor increases by one unit, while all other predictors are kept constant.

Continuing with the bakery example, the coefficient for the number of muffins (1.97) means that for every extra muffin ordered, the preparation time increases by about 1.97 minutes.

Thus, coefficients capture the **effect of each predictor** on the response.

Multiple Regression Example

Consider a **multiple regression** model with two predictors:

$$\text{Time} = 1.50 + 1.80x_1 - 0.75x_2$$

Here,

- x_1 = number of muffins
- x_2 = number of staff members working

We can interpret the intercepts and coefficients as follows:

- **Intercept (1.50)** → When there are zero muffins and no staff, the baseline preparation time is about 1.5 minutes
- **Coefficient of muffins (1.80)** → For every additional muffin ordered, keeping the number of staff constant, the preparation time increases by 1.8 minutes for every additional muffin
- **Coefficient of staff (-0.75)** → For each extra staff member, keeping the number of muffins constant, the preparation time decreases by 0.75 minutes for each additional member of staff

Therefore, if 6 muffins are ordered and 2 staff members are working,

$$\text{Time} = 1.50 + 1.80(6) - 0.75(2) = 1.50 + 10.80 - 1.50 = 10.80 \text{ mins}$$

This shows how **multiple predictors interact**: muffins increase preparation time, but extra staff reduces it.

Signs and Magnitude of Coefficients

Both the **sign** and the **magnitude** of coefficients are important.

Sign:

- A **positive coefficient** means that as the predictor increases, the response also increases
- A **negative coefficient** means that as the predictor increases, the response decreases

Magnitude:

- The **absolute value** of the coefficient shows the **strength** of the predictor's effect
- Larger values mean a stronger influence on the response

Handling Categorical Predictors

Not all predictors are numerical. Many are **categorical**, such as gender, region, or product type.

Since regression requires numerical input, categorical variables are converted into **dummy variables** (also called indicator variables).

1. Suppose we have a categorical variable "Region" with three levels: *North*, *South*, and *East*
2. We choose one level (say, *North*) as the **reference category**.
3. We then create two dummy variables, South and East
 1. South = 1 if the region is *South*, 0 otherwise
 2. East = 1 if the region is *East*, 0 otherwise
 3. Thus, if both South and East equal 0, we know the region is *North*

Now, the regression model includes these dummy predictors.

1. The coefficients of South and East show how much the response differs from the North region
2. The intercept captures the effect of the baseline category, North, while also accounting for other default effects included in the model

For example, if the coefficient for South is 4, it means that being in the South region increases the predicted outcome by 4 units compared to *North*, with all else being equal.

Evaluation Metrics

Evaluation metrics measure how well a regression model performs. They can be divided into explanatory metrics (how well the model explains the data) and predictive metrics (how accurate the predictions are). Predictive metrics can also be used as loss functions during training.

Explanatory Metrics

Explanatory metrics focus on how well a model **explains the relationships** in the data. These assess how well the model fits the data and how well it explains the variance within the feature space. By evaluating the explanatory metrics, we can answer questions like:

- Which features are significant?
- How strongly do they affect the outcome?
- Why does the outcome happen?

In other words, they answer: *“Does this model help us understand the data?”*

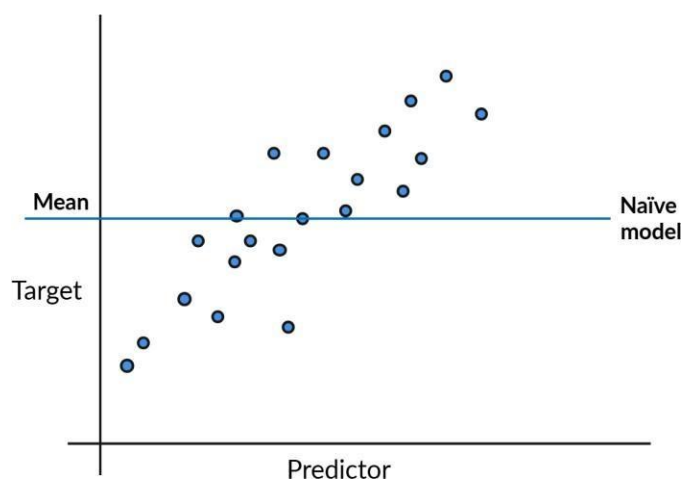
There are several explanatory metrics that we can use:

- **R^2 (Coefficient of Determination):** Proportion of variance explained
- **Adjusted R^2 :** Penalises for irrelevant predictors (better for explanatory regression)
- **p-values of coefficients:** Test whether predictors have significant effects
- **Confidence Intervals (CIs):** Provide uncertainty estimates around parameters

Let's focus on the one most commonly used, R^2 . Naturally, real world data will not have a linear relationship between the target feature and the predictors. There can only be a roughly linear relationship, with some variance in the data points. How well our model can capture this variance is an important factor while determining its usefulness in explaining the observed data.

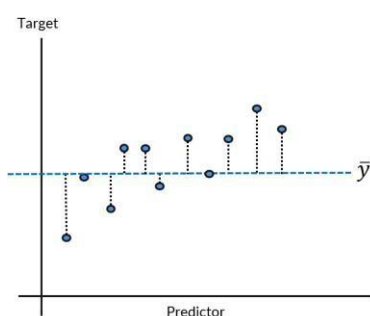
Now, note that R^2 is a relative measure. The best model will arguably be the one which best explains the variance of the data.

Similarly, the worst model will be the one which does not explain the variance at all – the baseline. This baseline, or naïve model can simply be a linear model having the value of the mean of the sample, $y = \bar{x}$.

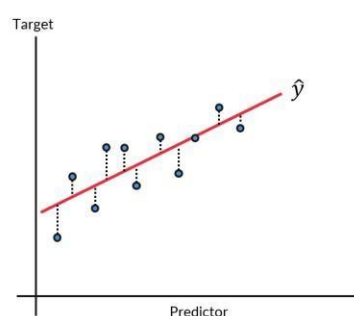


Now, in order for a model to be of any significance, it must be better at explaining the variance than this baseline model.

The following image shows this. The Residual Sum of Squares (RSS) should be less than the Total Sum of Squares (TSS).



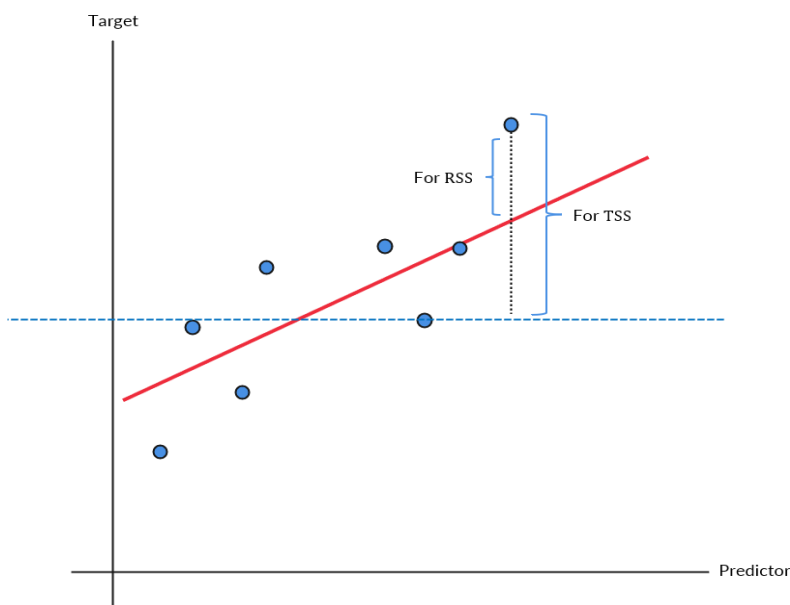
Total Sum of Squares



Residual Sum of Squares

TSS represents the deviation from the mean and RSS represents the deviation from the predictions. In other words, the variation of true values (y) around the significant model ($\hat{y}$) must be less than the variation of true values around the naïve model ($\bar{y}$).

We want the difference TSS – RSS to be higher.



We can define R^2 as the ratio of (TSS - RSS) to TSS,

$$R^2 = \frac{TSS - RSS}{TSS} = 1 - \frac{RSS}{TSS}$$

For a significant model, the maximum value RSS can take is equal to the TSS, and the minimum value is zero. Hence, the range of R^2 turns out to be $[0, 1]$.

- If $\frac{RSS}{TSS}$ is higher (close to 1) → model is more like baseline → R^2 decreases → less variance explained
- If $\frac{RSS}{TSS}$ is lower (closer to 0) → model is less like baseline → R^2 increases → more variance explained

An important aspect worth noting about R^2 is that it increases as we increase the number of input features. This is because more features will never reduce the model's ability to explain the variance in case of linear regression. However, this increase in variance does

not take into account whether or not the predictors are relevant to the model. We cannot unnecessarily keep adding less relevant features in order to achieve a better evaluation of the model. We need an appropriate trade-off between the increase in complexity and the gain in explained variance.

Worked Example: Interpreting R^2 in the Bakery Model

The bakery develops two linear regression models to predict preparation time for customer orders.

The evaluation results are shown below.

Metric	Model A	Model B
R^2	0.91	0.78

Analysis

The coefficient of determination (R^2) represents the proportion of variation in the response variable explained by the regression model.

For **Model A**,

$$R^2 = 0.91$$

This indicates that **91% of the variation in preparation time** can be explained by the number of muffins ordered.

The remaining **9%** of the variation may be due to factors that are not included in the model, such as:

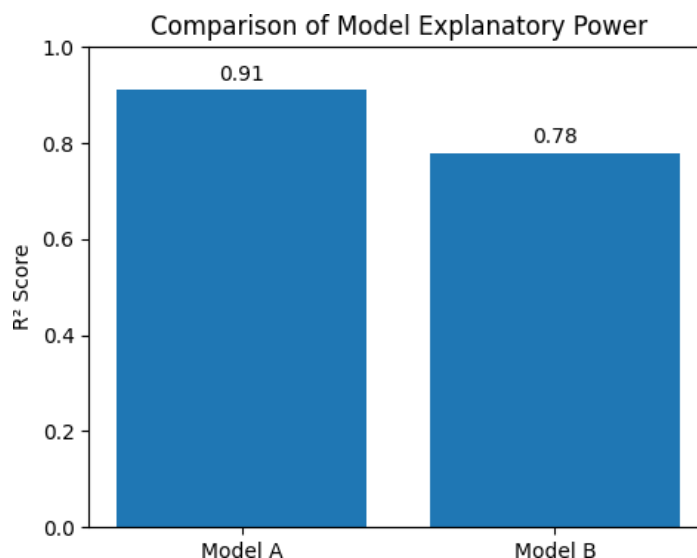
- Number of staff available
- Oven utilisation
- Complexity of customised orders
- Simultaneous customer requests

For **Model B**, only **78%** of the variation is explained by the predictor.

Therefore, **Model A provides a better explanation of the observed data.**

Business Interpretation

A higher R^2 suggests that the model captures a larger proportion of the observed variation. However, a high R^2 **does not necessarily imply that the model will perform well on unseen data**, particularly if unnecessary predictors have been included.



To account for this, we use the **adjusted coefficient of determination**, denoted as R^2_{adj} , which penalises the inclusion of unnecessary predictors.

$$R^2_{adj} = 1 - \frac{\frac{RSS}{n-p-1}}{\frac{TSS}{n-1}} = 1 - \frac{(1 - R^2)(n-1)}{n-p-1}$$

Here, n is the number of observations, and p is the number of predictors.

While R^2 rewards flexibility even when the improvement is artificial, adjusted R^2 corrects for this by introducing a penalty term based on **degrees of freedom**.

In regression, each predictor you add “uses up” one degree of freedom. That means you are spending some of your information capacity just to estimate another parameter.

- If the new predictor explains a genuine signal, the tradeoff is worth it
- If it does not, then you’ve wasted a degree of freedom, and adjusted R^2 reflects this by going down

Predictive Metrics

While R^2 and adjusted R^2 provide insight into how much variance is explained by the model, they primarily assess goodness of fit on the training data. These measures are not always reliable indicators of how well the model will perform on unseen data, particularly if overfitting has occurred.

Why Adjusted R^2 ?

Adding more predictors to a regression model almost always increases (or at least does not decrease) the R^2 value, even if the additional predictors contribute very little meaningful information.

Adjusted R^2 addresses this limitation by introducing a penalty for unnecessary predictors. Unlike R^2 , Adjusted R^2 increases **only when a newly added predictor improves the model sufficiently to justify its inclusion**.

Consequently, Adjusted R^2 is generally preferred when comparing multiple regression models containing different numbers of predictors.

Formula

$$\text{Adjusted } R^2 = 1 - (1 - R^2) \left(\frac{n - 1}{n - p - 1} \right)$$

where

- n = Number of observations
- p = Number of predictors

Solved Example

The bakery analytics team evaluates two competing multiple linear regression models.

Metric	Model A	Model B
R^2	0.95	0.94
Adjusted R^2	0.86	0.92

Although Model A has a slightly higher R^2 , its Adjusted R^2 is substantially lower.

This indicates that several predictors contribute little useful information and unnecessarily increase model complexity.

Therefore, **Model B is preferred**, as it provides a better balance between explanatory power and model simplicity.

When developing predictive models for business applications, analysts should avoid selecting a model solely because it has the highest R^2 value. Models with many irrelevant predictors may appear to perform well on the training data but often generalise poorly. Adjusted R^2 provides a more reliable basis for comparing regression models of different complexity.

Practice Exercise

A data scientist develops three regression models for predicting bakery preparation time.

Model	R^2	Adjusted R^2
A	0.96	0.81
B	0.93	0.91
C	0.89	0.88

Tasks

1. Which model would you recommend for deployment? Justify your answer.
2. Explain why Model A has the highest R^2 but the lowest Adjusted R^2 .
3. Under what circumstances would R^2 alone be sufficient for evaluating a regression model?

To properly evaluate a model's **predictive ability**, we use error-based predictive metrics. These metrics directly quantify the discrepancy between the observed values (y_i) and the predicted values ($\hat{y}_i$).

These models not only tell how accurate the model predictions are, but also suggest how well it generalises the learnt patterns on unseen data.

Now, as we have already discussed earlier while talking about loss functions, we need a single measure of error. We cannot use the error for a single data point to evaluate the whole model. Many such aggregate measures are there which give insight into the predictive performance of the model.

Mean Squared Error (MSE)

MSE is simply the mean of the square of individual residuals, or the mean of RSS.

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

It penalises larger errors more heavily due to squaring and proves useful when large deviations are particularly undesirable.

The scale of MSE is in square of units of y . We often compare MSE with the target's variance, as both measure the average of squared deviations. A model is said to perform particularly well if $\text{MSE} \ll \text{variance}$.

Root Mean Squared Error (RMSE)

RMSE is the square root of the MSE:

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

It retains the same penalisation of larger errors as MSE due to squaring but returns to the original scale of the target variable y , making interpretation more intuitive.

RMSE is widely used when we care about large deviations being especially costly, and its unit consistency with y makes it preferable for reporting absolute error magnitudes.

Mean Absolute Error (MAE)

MAE is the mean of the absolute differences between predictions and actual values:

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

Unlike MSE and RMSE, it does not square the errors; hence, it penalises deviations linearly. This makes MAE more robust to outliers and easier to interpret, as it corresponds to the average absolute deviation in the units of y . However, it may underemphasise large errors compared to RMSE.

Mean Absolute Percentage Error (MAPE)

MAPE expresses prediction errors as percentages of the actual values:

$$\text{MAPE} = \frac{100}{n} \sum_{i=1}^n \left| \frac{y_i - \hat{y}_i}{y_i} \right|$$

It provides a scale-independent measure, making it particularly useful for comparing models across datasets with different units or magnitudes.

MAPE is easily interpretable: e.g., a MAPE of 10% means predictions are, on average, 10% away from the true values. However, it has limitations: it is undefined when $y_i = 0$ and can be disproportionately large when actual values are very close to zero.

Final Model Evaluation

Building a regression model is only one part of the machine learning workflow. Before deploying the model for business use, it is important to evaluate whether the model is accurate, reliable, and capable of making predictions on unseen data.

A good regression model should satisfy the following criteria:

Evaluation Aspect	Question to Ask
Prediction Accuracy	Are the predicted values close to the observed values?
Goodness of Fit	Does the model explain a substantial proportion of the variation in the data?
Generalisation	Does the model perform consistently on new data?
Interpretability	Can the influence of each predictor be explained?

Model evaluation should therefore combine multiple metrics rather than relying on a single indicator.

Case Study: Selecting a Production Model

The bakery analytics team develops two candidate regression models before deploying the prediction system.

The evaluation results are summarised below.

Metric	Model A	Model B
MSE	2.35	1.84
R ²	0.89	0.92
Adjusted R ²	0.87	0.91

Analysis

Model B achieves:

- Lower prediction error (MSE)
- Higher explanatory power (R²)
- Better adjusted explanatory performance (Adjusted R²)

Although both models perform well, Model B consistently outperforms Model A across all evaluation metrics.

Consequently, the bakery selects **Model B** for deployment.

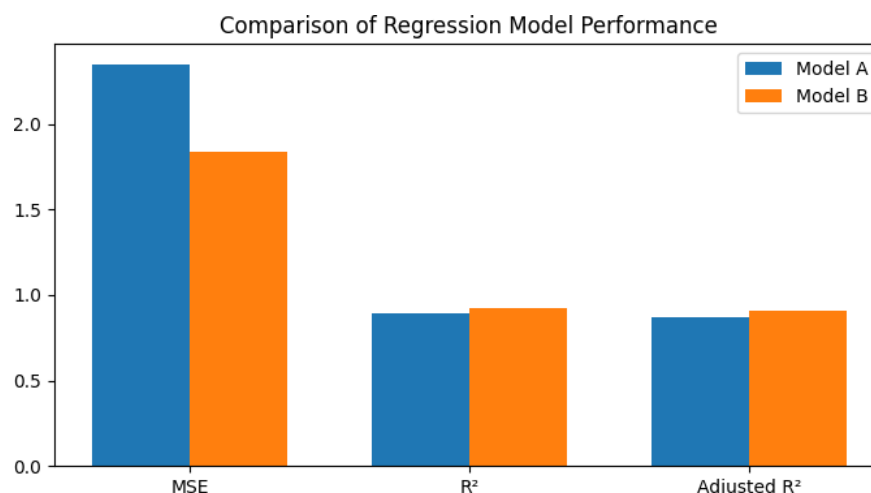
Deployment Considerations

Even after a model performs well during evaluation, organisations should continue monitoring its performance after deployment.

Some practical considerations include:

- Changes in customer ordering patterns.
- Seasonal demand variations.
- Changes in staffing levels.
- Introduction of new bakery products.

If these conditions change significantly, the regression model should be retrained using updated historical data.



Practice Exercise

The bakery is opening a third outlet.

The operations team has collected six months of historical preparation-time data and developed two regression models.

You have been asked to recommend one model for deployment.

Before making your recommendation:

1. Identify the evaluation metrics you would examine.
2. Explain why relying only on R^2 may lead to an incorrect conclusion.
3. Describe two situations in which the model should be retrained after deployment.
4. Discuss how business requirements, in addition to statistical metrics, can influence model selection.

Although Linear Regression is one of the oldest machine learning algorithms, it continues to be widely used in finance, healthcare, manufacturing, marketing, and operations research. In many organisations, Linear Regression serves as the baseline model against which more sophisticated machine learning algorithms are compared because of its simplicity, interpretability, and computational efficiency.

Ordinary Least Squares (OLS)

One natural question arises after defining the Mean Squared Error (MSE): **how does a regression model determine the best values of its coefficients?**

Rather than selecting the intercept and slope manually, linear regression estimates the values that minimise the prediction error across all observations in the training dataset.

This optimisation approach is known as **Ordinary Least Squares (OLS)**.

OLS determines the regression coefficients by identifying the line for which the **Residual Sum of Squares (RSS)** is the smallest.

Mathematically,

$$RSS = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

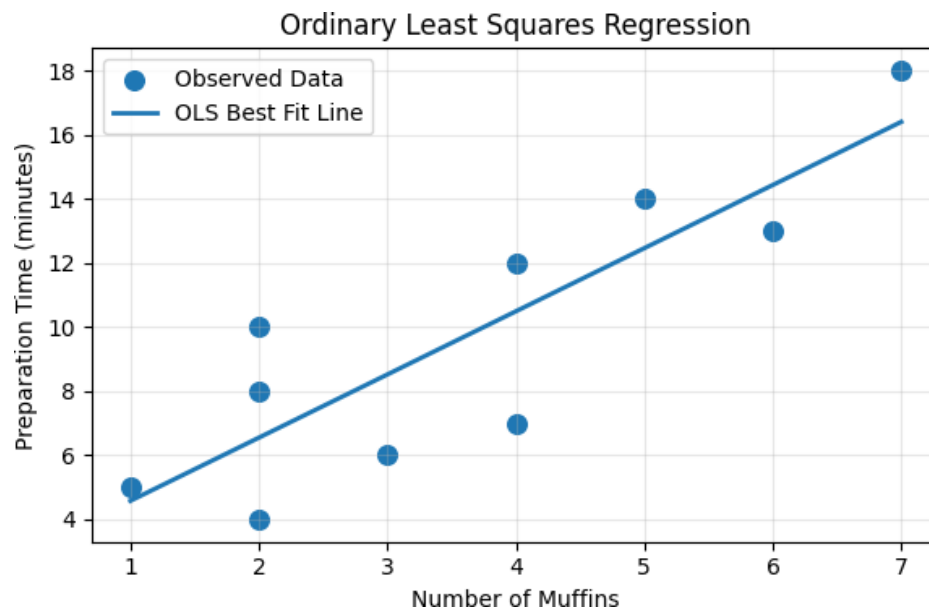
The optimal regression model is therefore the one that minimises this quantity.

Unlike manually choosing a regression line, OLS guarantees that the fitted coefficients correspond to the minimum possible squared prediction error for the observed data.

Inference and Interpretation

Up to this point, we have mostly treated linear regression as a predictive tool: input data, obtain fitted coefficients, and then evaluate performance. We also know how to interpret coefficients and their effects. But how do we ensure the significance of these coefficients and reliability of the results.

Regression is also a powerful **inferential tool**: it allows us to ask questions about the strength, reliability, and significance of relationships between predictors and outcomes. And one of the most basic ways to quantify the reliability of coefficients is through **confidence intervals**. The confidence intervals for a coefficient provide a range of plausible values for the parameter estimate.



Ordinary Least Squares searches for the regression line that minimises the overall squared prediction error.

Solved Example: Why OLS Selects One Model over Another

The bakery analytics team evaluates two candidate regression models using the same historical dataset.

Model	RSS
Model A	22.8
Model B	15.3

Since Ordinary Least Squares selects the model that minimises the Residual Sum of Squares, **Model B** is preferred.

Although both models capture the general trend, Model B produces predictions that are collectively closer to the observed preparation times.

Consequently, the regression coefficients associated with Model B are considered the optimal coefficients.

Interpretation: OLS does not seek the line that passes through every observation. Instead, it identifies the line that minimises the overall squared prediction error across all observations.

Practice Exercise

The bakery develops three different regression models.

Model	RSS
A	18.6
B	11.9
C	24.7

Without performing any additional calculations, answer the following:

1. Which model would Ordinary Least Squares select?
2. Explain why minimising RSS generally improves prediction quality.
3. Why is RSS preferred over simply summing prediction errors?

Gradient Descent

Ordinary Least Squares (OLS) provides a direct mathematical solution for estimating the regression coefficients. However, as the number of predictors increases or when working with very large datasets, directly solving the normal equation can become computationally expensive.

In such situations, regression models are commonly trained using **Gradient Descent**, an iterative optimisation algorithm.

Instead of computing the optimal coefficients in a single step, Gradient Descent begins with an initial estimate and gradually updates the coefficients to minimise the prediction error.

The objective is to minimise the Mean Squared Error (MSE) by repeatedly adjusting the regression coefficients in the direction that reduces the loss.

Mathematical Update Rule

The coefficient update at every iteration is given by

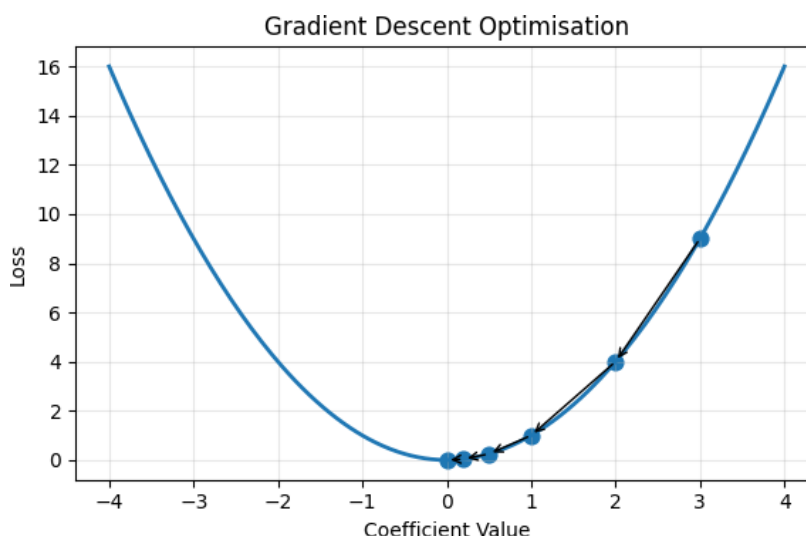
$$\theta = \theta - \alpha \frac{\partial J(\theta)}{\partial \theta}$$

where

- θ represents the model coefficient
- α is the learning rate
- $J(\theta)$ is the loss function (typically MSE)

Each iteration moves the model parameters towards the minimum of the loss function.

Gradient Descent gradually updates the regression coefficients until the minimum loss is reached.



Solved Example

Selecting an Appropriate Learning Rate

The bakery analytics team trains the preparation-time prediction model using Gradient Descent.

Three different learning rates are evaluated.

Learning Rate	Observation
1.0	Training diverges and the loss increases.
0.1	Training converges quickly to a stable solution.
0.0001	Training converges very slowly and requires many iterations.

Analysis

The learning rate determines the size of each optimisation step.

- A **very large learning rate** may overshoot the optimum, preventing convergence.
- A **very small learning rate** converges slowly, increasing training time.

- An appropriately chosen learning rate provides stable and efficient optimisation.

For this problem, a learning rate of 0.1 offers the best balance between convergence speed and stability.

Professional Insight

In practical machine learning workflows, selecting an appropriate learning rate is one of the most important hyperparameter tuning decisions. Modern optimisation algorithms such as **Adam**, **RMSProp**, and **AdaGrad** automatically adapt the learning rate during training, making optimisation more efficient for complex models.

Practice Exercise

A bakery develops a regression model using Gradient Descent.

During training, the loss values observed over successive iterations are shown below.

Iteration	Loss
1	84.2
10	37.5
25	15.8
50	8.7
100	8.5

Tasks

1. Explain whether the optimisation process appears to be converging.
2. Identify the point after which further improvement becomes marginal.
3. Suggest one reason why the loss may stop decreasing after many iterations.
4. Recommend one strategy to improve optimisation if convergence becomes unstable.

Confidence Intervals for Coefficients

When we estimate the coefficients for a feature, we are working with a sample data rather than the entire population. While a point estimate, β_j , gives us a single best guess for the effect of a predictor, it does not convey how precise that estimate is.

This means our estimates, $\hat{\beta}_j$, are subject to uncertainty. Confidence Intervals (CIs) provide a way to quantify that uncertainty and ask: *how sure are we about the value of this coefficient?*

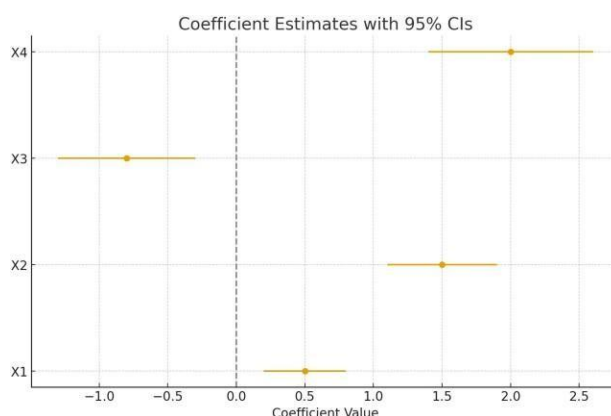
In regression, a 95% confidence interval for β_j is constructed. So, if we were to repeatedly sample data and refit the model, about 95% of those intervals would contain the true parameter.

For example, if the CI for a coefficient is [1.2, 2.8], we interpret this as evidence that the true effect of that predictor is likely somewhere between 1.2 and 2.8.

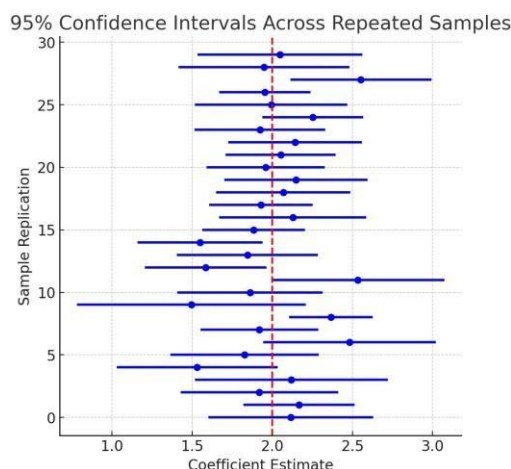
The width of the interval carries important information:

- **Narrow intervals** mean more stable estimates and greater confidence in the reported effect size
- **Wide intervals** suggest higher uncertainty, which may arise from limited data, high variance, or multicollinearity among predictors

Consider the two plots below.



This plot shows the estimated coefficients and their 95% confidence intervals for different predictors - if the confidence interval for a predictor includes the coefficient value of 0, there is a possibility that the predictor's coefficient can be 0, and the predictor might not be significant.



This plot shows the estimated coefficient with confidence intervals for different samples of the same data. The red line shows the actual coefficient value. So, for different samples, the estimated coefficient may come out to be different.

A 95% confidence interval (CI) for a coefficient β_j is constructed as:

$$CI = \hat{\beta}_j \pm t_{\alpha/2, df} \cdot SE(\hat{\beta}_j)$$

where $SE(\hat{\beta}_j)$ is the standard error of the estimate, and α is the level of significance (0.05).

Prediction Intervals

While confidence intervals tell us how precisely we have estimated the mean effect of predictors, prediction intervals give a range where we expect a new individual observation to fall.

Even if we know the regression line perfectly, individual outcomes vary due to randomness (noise). Prediction intervals capture this uncertainty, for a particular prediction.

On the other hand, confidence intervals for the **mean response** of Y tell us how confident we are while predicting $\hat{Y}$ at a certain point X . The mean response refers to uncertainty in estimating the regression line itself.

If we predict $\hat{Y}$ over and over again for the same value of X in different samples of the data, we get a range of $\hat{Y}$ values which give us the mean response at that X .

Obviously, we will have more points lying near the mean of X and less at the extremes. Hence, we can pin down the mean response near the mean of the sample better than at the extremes. This means, confidence intervals are narrower near the mean of the sample and wider at the extremes.

PI are **wider than** CI for the mean response because they include both:

- Uncertainty in estimating the regression line
- Variability of individual responses

So, PI might also be **narrower in the centre** (where most data lie) and **wider at extremes** (where data are sparse).

Consider this linear model relating study hours(X) to exam score(Y):

$$\hat{Y} = 40 + 5X$$

Predicted score for a student who studies 5 hours:

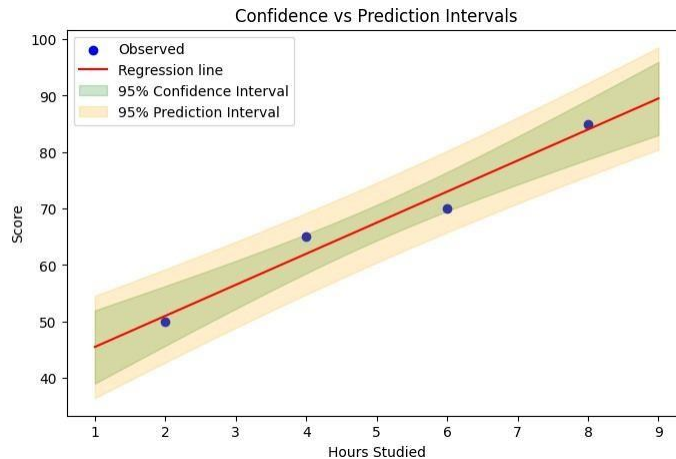
$$\hat{Y}_0 = 40 + 5 \cdot 5 = 65$$

Suppose the standard error for prediction at $X = 5$ is $SE_{\text{pred}} = 5$. 95%

Prediction Interval:

$$\hat{Y}_0 \pm t_{0.025, df} \cdot SE_{\text{pred}} \approx 65 \pm 2 \cdot 5 = [55, 75]$$

Interpretation: If a student studies 5 hours, we are 95% confident that their actual score will fall between 55 and 75.



The distance between CI and PI shows the extra uncertainty from individual variability, and both bands reflect how much data support is available in different X regions.

Once we have explored confidence and prediction intervals, we can take the next step: testing whether a predictor truly influences the outcome. This is done using **hypothesis tests** for the regression coefficients. In simple terms, we test if the effect we see in our sample is likely to exist in the population or has it occurred by chance.

In practice, we calculate a test statistic for each coefficient and compare it to a standard reference (or use a p-value). If the result shows a strong enough signal, we conclude that the predictor is **statistically significant**; otherwise, we consider it inconclusive. This is along the same lines as confidence intervals: a coefficient whose interval does not include zero is also considered significant.

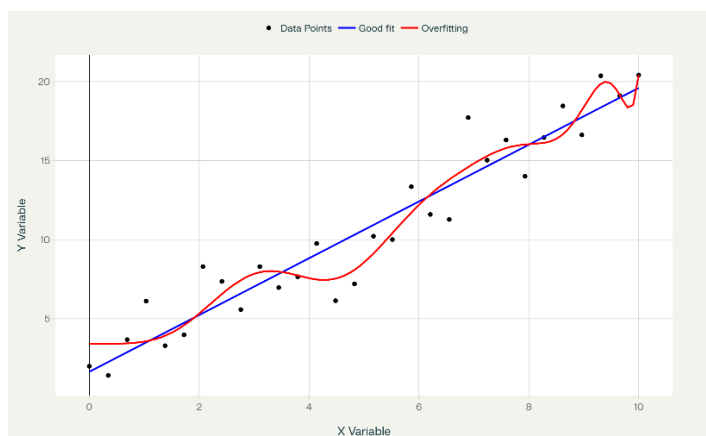
Challenges in Linear Regression

While regression is a powerful tool to understand relationships and make predictions, real-world data rarely behaves perfectly. Several challenges can arise that affect the reliability and interpretability of our models. Understanding these issues helps us diagnose problems and take corrective measures.

Overfitting

Overfitting happens when a model captures not only the underlying pattern but also the random noise in the data. In linear regression, this can occur if we include too many predictors relative to the number of observations, or when we try to fit very complex relationships with limited data. A good model should be able to **generalise** its learnings from training to new and unseen data as well.

We can notice overfitting when a model predicts almost perfectly on the training data but performs poorly on new, unseen data. Overfitted models tend to give overly optimistic confidence and prediction intervals, making them unreliable.

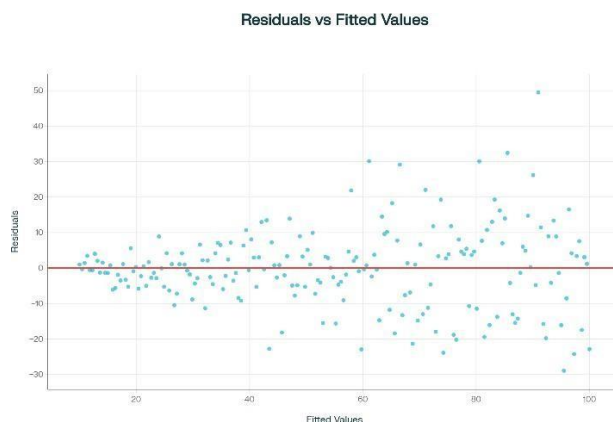


The solution is to simplify the model — keep only the most relevant predictors, combine similar variables, or limit how many iterations the optimiser (like gradient descent) does during training.

We can also use advanced **regularisation** techniques. These adjustments help the model focus on real signals rather than noise.

Heteroscedasticity

Regression assumes that the variability of errors is roughly the same across all levels of the predictors. When this is not true, we have heteroscedasticity. For example, the spread of residuals might increase as the predicted values grow larger, forming a funnel-like shape in a residuals vs fitted plot.



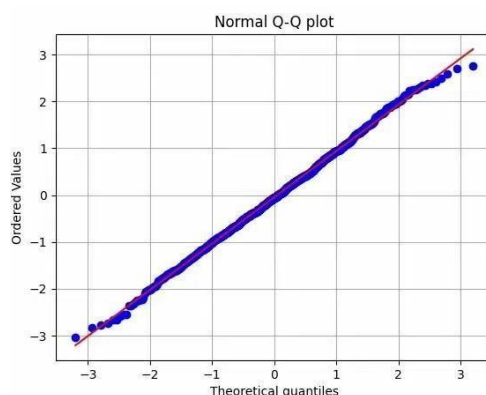
Heteroscedasticity affects the accuracy of standard errors, which in turn can make confidence intervals and hypothesis tests misleading. To address this, one might **transform the response variable** (e.g., take a log or square root) to bring the target feature and error on expected scale.

Non-Normality of Errors

While the regression line itself can still be a valid estimate when residuals are not perfectly normal, inference becomes tricky. Non-normal residuals can be skewed, heavy-tailed, or show systematic patterns.

We can detect this using **histograms** or **quantile-quantile (QQ) plots**, where the residual quantiles are compared against theoretical normal quantiles.

Quantiles are points that divide your data into equal proportions. For example, the 25th percentile (first quartile) is the value below which 25% of observations fall.



QQ-plots visually check quantiles: if residuals are normal, the actual quantiles match the output quantiles of the sample and the points lie roughly along a straight 45° line. Any deviations of sample quantiles from theoretical ones indicate non-normality.

If the points deviate noticeably from a straight line, the errors are not normally distributed.

Non-normality mainly affects confidence intervals and hypothesis tests: they may no longer be accurate. A common fix is to **transform the response variable** or, if necessary, use **non-parametric methods** that do not assume normality.

Multicollinearity

Multicollinearity arises when predictors are highly correlated with each other. In such cases, it becomes difficult to isolate the individual effect of a predictor on the response, because changes in one predictor tend to coincide with changes in others.

We can detect multicollinearity in several ways. One intuitive sign is when adding or removing a variable causes unusually large changes in the estimated coefficients. A more formal measure is the **Variance Inflation Factor (VIF)**, which is calculated for each predictor X_j as:

$$VIF_j = \frac{1}{1 - R_j^2}$$

Here, R_j^2 is the R^2 obtained when regressing X_j on all the other predictors. A VIF of 1 means no correlation with other predictors, while values above 5 or 10 indicate strong multicollinearity.

The larger the VIF, the more the variance of the coefficient β_j is 'inflated' due to correlation with other predictors. High VIFs imply unstable coefficient estimates, wide confidence intervals, and less reliable hypothesis tests.

To mitigate multicollinearity, we can:

- Remove or combine correlated predictors
- Perform recursive feature elimination (RFE) to check removing which features fixes multicollinearity
- Apply **regularisation methods** that affect coefficients and stabilise estimate

